

GEOG 3319: Geographic Field Techniques

Lecture 09 Survey123 and ArcGIS Collector

Yaqian He, PhD
Assistant Professor
Department of Geography



Geography

Survey 123

Overview

- Survey123
- Authoring surveys
- Collection data
- Decision

○ Forms

A printed or typed document with blank spaces for insertion of required information

✓ Universally used

- Utilities (Water, Electric, Telco, Gas)
- Transportation
- Government
- Humanitarian, Non-Profit, Conservation
- Census
- Health
- Emergency Management
- Agriculture

UTILITY SLIP		
NAME _____		
(Check proper number)		
1. Admit to	Library	Study Hall Classroom
2. Tardy.		
3. Not Tardy.		
4. Excused.		
5. Not Excused.		
6. Detained by	_____	
7. Report at office	_____	
8. Phone call No.	_____	
9. Call home.	_____	
10.	_____	
Attention of _____		
Date _____		
Hour _____		
Issued by _____		
Form 23	School Service, Inc.	(800) 747-9549

Suvery123

- Form- centric data collection solution
- Leverages smart forms
- Web browser-based solution
 - ✓ Also supports mobile devices
- Integrated with ArcGIS online
 - ✓ Available with ArcGIS Enterprise



Why Smart Forms in ArcGIS

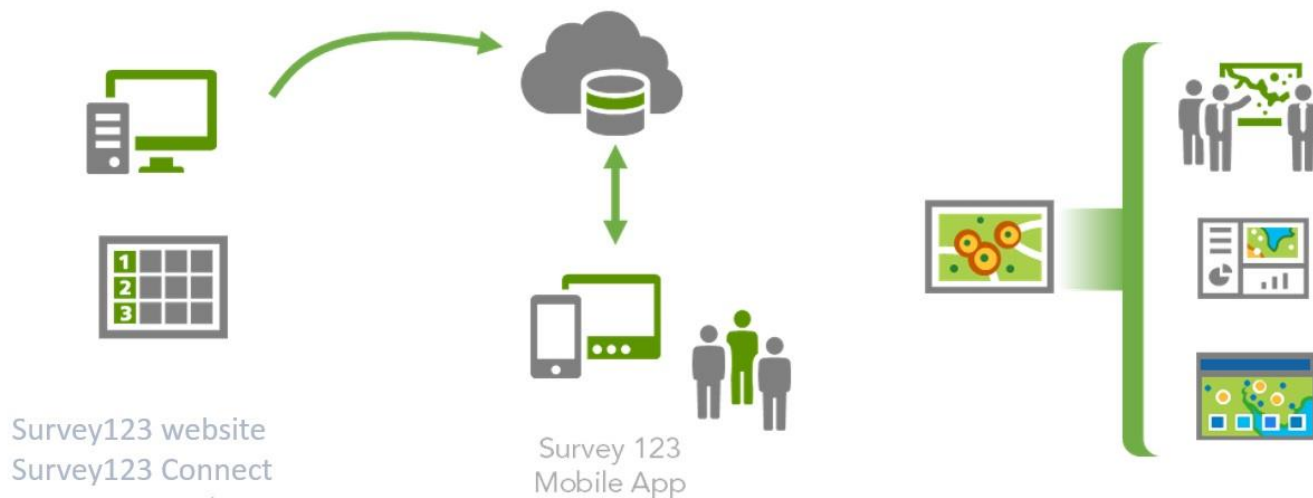
- Reduce errors
 - ✓ During data capture
 - ✓ Transcribing data from paper to digital
- Boost field data capture productivity
 - ✓ Precompute responses
 - ✓ Associate photos with alphanumeric data
- Data from the field to the office in real-time
- All data captured is georeferenced



What makes a form 'Smart'?

- Types of questions
 - ✓ Text, Integers, Dates, Time, Signatures, Photos, Scales
 - ✓ Multiple choices, Single choice lists, Barcode, Sketch
- Form logic and validation
 - ✓ Skips, Cascading selects, Default Expressions
 - ✓ Precomputed responses, Mandatory questions, Favorite answers
 - ✓ Use reference data
- Look and feel
 - ✓ Multiple-language support, Themes, Hints

Suvery123



Authoring Surveys

- Survey123 Web Designer
 - ✓ Build smart forms graphically in a web browser
 - ✓ Very easy to get started
 - ✓ Ideal for simple forms
- Survey123 Connect
 - ✓ Desktop tool, works in combination with Microsoft Excel
 - ✓ Requires learning and familiarity with XLSForms specification

Authoring Surveys

Create a New Survey

Using the web designer

- Get started quickly
- Best for simple surveys
- Author your survey graphically



Get started

Using Survey123 Connect

- Using a desktop application
- Full smart form capabilities
- Author through XLSForm spreadsheet



Get started

Authoring Surveys

- Preview in left panel, set parameters in the right panel

ArcGIS Survey123

My Surveys

Help

Yaqian.He@Dartmouth.edu

UCA vegetation mapping

OverviewDesignCollaborateAnalyzeDataSettings

UCA vegetation mapping

UCA campus vegetation mapping using Survey123

1

Deciduous or Evergreen?

-Please Select-

2

Condition

-Please Select-

3

Height

In Meters

123

+

Add

Edit

Appearance

Settings

Singleline Text

Multiline Text

Single Choice

Single Choice Grid

Multiple Choice

Dropdown

Rating

Likert

Number

Date

Time

Date/Time

Image

File Upload

Map

Ranking

Saved

Preview

Publish

Authoring Surveys

○ Survey Question Types in Web Designer

- ✓ Most common questions
- ✓ Addresses many workflows
- ✓ Simple forms

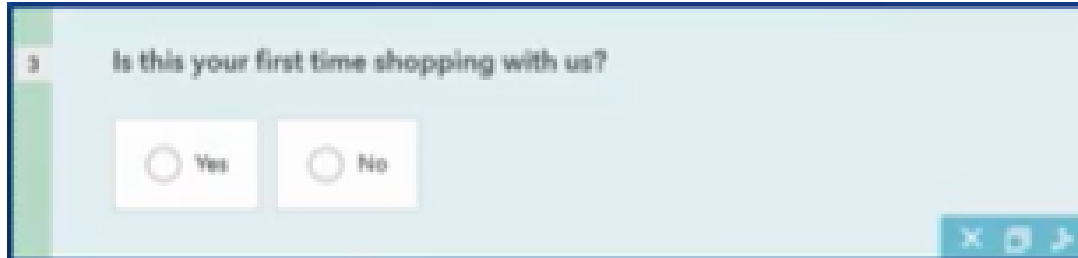
 Email	 Website
 Signature	 Note
 Group	 Page

 Add	 Edit	 Appearance	 Settings
 Singleline Text	 Multiline Text		
 Single Choice	 Single Choice Grid		
 Multiple Choice	 Dropdown		
 Rating	 Likert		
 Number	 Date		
 Time	 Date/Time		
 Image	 File Upload		
 Map	 Ranking 12		

Authoring Surveys

○ Setting Rules

- ✓ Simple conditional statements can control visibility of questions
- ✓ Applicable for Single choice, Dropdown, and Rating...

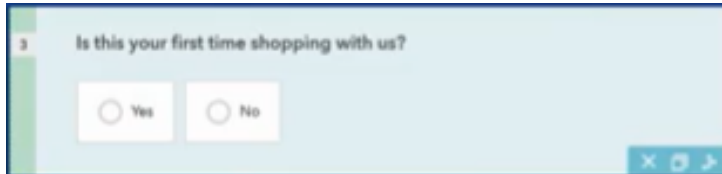


A screenshot of a survey interface. The question is "Is this your first time shopping with us?". Below the question are two radio button options: "Yes" and "No". The "Yes" option is selected. The interface has a light blue background and a dark blue border. There are small icons in the top left and bottom right corners of the survey area.

Authoring Surveys

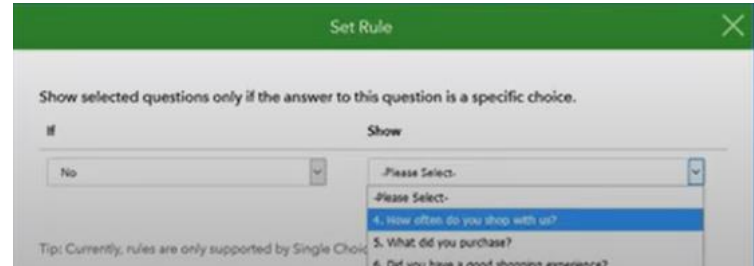
○ Setting Rules

- ✓ Simple conditional statements can control visibility of questions
- ✓ Applicable for Single choice, Dropdown, Likert, and Rating



3 Is this your first time shopping with us?

☐ Yes ☐ No



Set Rule

Show selected questions only if the answer to this question is a specific choice.

If	Show
No	Please Select- Please Select- 4. How often do you shop with us? 5. What did you purchase? 6. Did you have a good shopping experience?

Tip: Currently, rules are only supported by Single Choice

Authoring Surveys

○ Style Your Form

ArcGIS Survey123 ▾ My Surveys Help

Yaqian.He@Dartmouth.edu ▾

River Study

Overview **Design** Collaborate Analyze Data Settings

River Study

River Study using Survey123

1

Site Name

☐ Site 1

☐ Site 2

☐ Site 3

☐ Site 4

☐ Site 5

☐ Site 6

2

Chanel Depth
In Meters

Add

Edit

Appearance

Settings

Layout Element

Survey header

Survey description

Survey footer

Theme

Save

Preview

Publish

Authoring Surveys

○ Updating Surveys

- ✓ Published surveys can be edited in Web Designer
- ✓ Supported changes
 - Add new questions
 - Reorder questions
 - Delete questions
 - Update labels in questions
 - Add, update, delete choices in lists
 - Update constraints, default values, hints, and required status

Sharing Surveys

○ Private Surveys

- ✓ Shared with a well-known group of users (secured access)
- ✓ Requires use of one ArcGIS accounts per every filed user
- ✓ Enables Editor Tracking capabilities
- ✓ Ideal for sensitive data or when you want to understand who is doing what

○ Public Surveys

- ✓ Shared with everyone
- ✓ No need for ArcGIS accounts; anyone can submit data
- ✓ Ideal for crowd-sourcing initiatives

Sharing Surveys

○ Collaborate Tab

✓ After publishing survey, you can control access

- Data submission → submitter (share survey)
- Review results → Viewer (share results)

ArcGIS Survey123 ▾ My Surveys Help Yaqian ▾

UCA tree mapping_YH Overview Design **Collaborate** Analyze Data Settings

Share survey
Share results
Update survey
Group settings

Share this survey

Link

☒ Open the survey in browser directly

☐ Ask the user how to open the survey, in browser or in the Survey123 field app

☐ Open the survey in the Survey123 field app directly. ([Learn more about this option](#))

Embed

[Embed in website](#)

Who can submit to this survey?

☒ Everyone (Public)

☒ Members of my organization(University of Central Arkansas)

☐ Following groups:

[Details](#)

Control submission

○ Collaborate Tab

- ✓ After publishing survey, you can control access
 - Survey Status: open/close

ArcGIS Survey123 ▾ My Surveys Help Yaqian ▾

UCA tree mapping_YH Overview Design **Collaborate** Analyze Data Settings

Share survey

Share results

Update survey

Group settings

Multiple submissions ⓘ

☐ Enable draft mode

☒ Allow multiple submissions

☐ Only one submission is allowed per submitter ⓘ

Survey status ⓘ

This survey is currently open.

☒ Open: accepting responses

☐ Closed: not accepting responses

☐ Schedule the open/close date and time (based on -05:00 time zone)

Open: ⓘ ⓘ

Close: ⓘ ⓘ

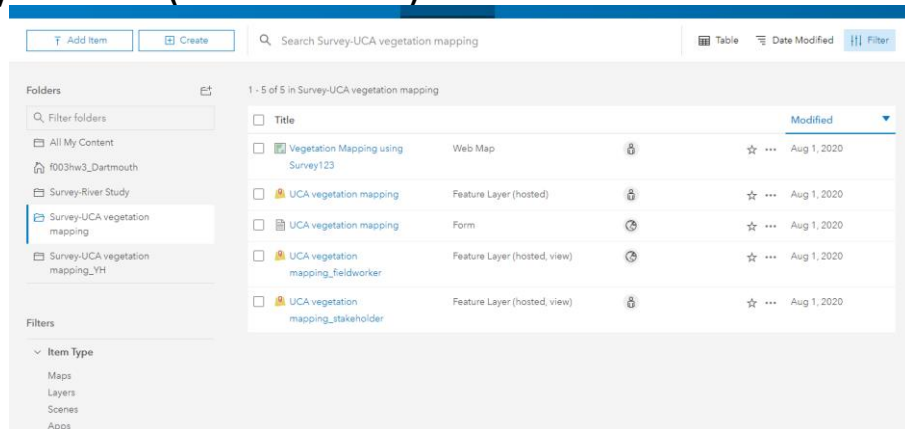
Save

Surveys in ArcGIS online

- A survey is stored as a folder

- ✓ May have 4 components

- Form
- Feature layer
- Feature layer view (Fieldworker)
- Feature layer view (stakeholder)



Accessing the Survey

- Control how the survey is accessed
 - ✓ Link to share the survey

Share this survey

Link

<https://arcg.is/SHfGS>



- ☒ Open the survey in browser directly
- ☐ Ask the user how to open the survey, in browser or in the Survey123 field app
- ☐ Open the survey in the Survey123 field app directly. ([Learn more about this option](#))

Embed

[Embed in website](#)

Accessing the Survey

○ Control how the survey is accessed

- ✓ Link to share the survey
- ✓ Embed in a web page

Embed in website

Copy and paste HTML to embed in website. [Copy](#)

```
<style>.embed-container {position: relative; height: 0; padding-bottom: 80%; max-width: 100%;} .embed-container iframe, .embed-container object, .embed-container iframe[position: absolute; top: 0; left: 0; width: 100%; height: 100%;] small {position: absolute; z-index: 40; bottom: 0; margin-bottom: -15px;}</style><div class="embed-container"><iframe name="survey123webform" width="500" height="400" frameborder="0" marginheight="0">
```

Embed options

☒

Show options bar (sign in info, multiple language options, etc.)

☒

Show survey header

☒

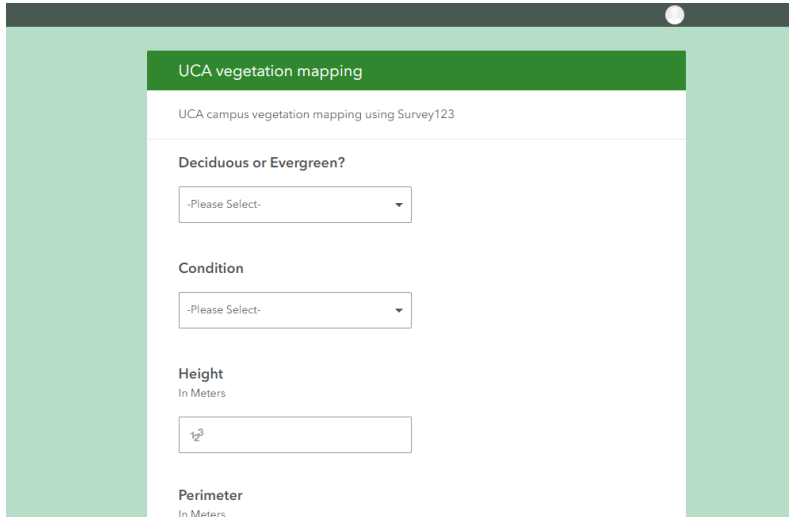
Show survey description

OK

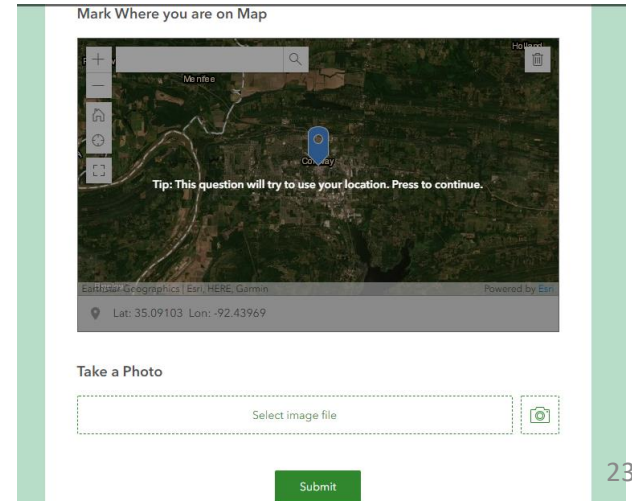
Collect data

○ Web forms

- ✓ Capture data from a web browser
- ✓ Can be embedded within a website
- ✓ Add and edit data



The screenshot shows a web form titled "UCA vegetation mapping" with a subtitle "UCA campus vegetation mapping using Survey123". The form contains three sections: "Deciduous or Evergreen?" with a dropdown menu showing "-Please Select-"; "Condition" with a dropdown menu showing "-Please Select-"; and "Height" with a text input field containing "12". Below these is a "Perimeter" label with "In Meters" underneath it.



The screenshot shows a map interface titled "Mark Where you are on Map". It features a map with a blue location pin and a search bar. Below the map, there is a "Take a Photo" section with a dashed box for "Select image file" and a camera icon. At the bottom, there is a green "Submit" button. The map also displays coordinates: "Lat: 35.09103 Lon: -92.43969".

Collect data

○ Mobile App

- ✓ Use for data collection in the field
- ✓ Can capture new data, edit existing data
- ✓ Supports offline workflows
- ✓ Cross platform

The screenshot shows a mobile app interface for "UCA vegetation mapping". The status bar at the top indicates AT&T service, 6:14 PM, and 15% battery. The app header is green with a close button (X), the title "UCA vegetation mapping", a signal strength icon, and a menu icon (three horizontal lines). Below the header, the text "UCA campus vegetation mapping using Survey123" is displayed. The form contains several fields: "Deciduous or Evergreen?" with a dropdown menu; "Condition" with a dropdown menu; "Height" with a label "In Meters" and a text input field; "Perimeter" with a label "In Meters" and a text input field; "Species" with a dropdown menu; and "Mark Where you are on Map" with a location pin icon, the coordinates "43°41'N 72°17'W ± 5 m", and a close button (X). At the bottom right, there is a green bar with a white checkmark icon.

UCA campus vegetation mapping using Survey123

Deciduous or Evergreen?

Condition

Height
In Meters

Perimeter
In Meters

Species

Mark Where you are on Map
43°41'N 72°17'W ± 5 m

Collect data

- GNSS device direct connection
 - ✓ Support for high accuracy collection
 - ✓ Store metadata to understand quality
 - ✓ Build validation rules for execution in the field

Collect data

○ External devices

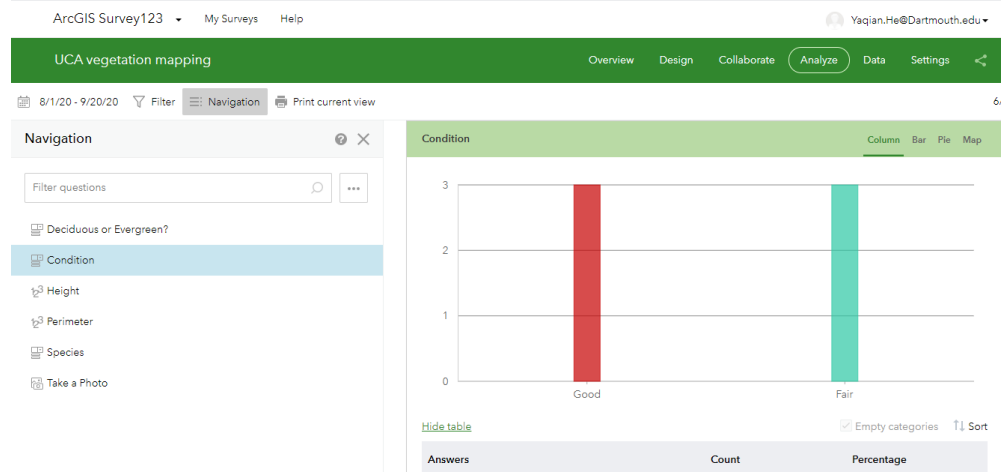
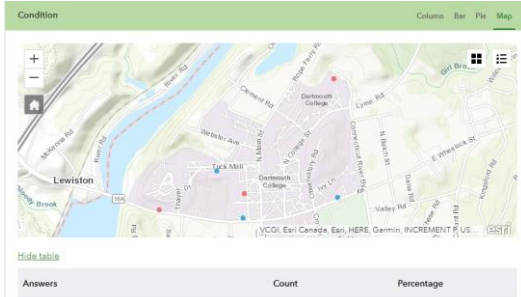
- ✓ Rangefinder
- ✓ Measure objects:
 - Distance
 - Location
 - Length/ Area



Spike, ike GPS

Make decision

- Ready to use reporting tools
 - ✓ Understand data submission patterns
 - ✓ Create detailed reports
 - ✓ Aggregate survey data
 - ✓ Download data



Make decision

- Feature reports

- ✓ Create printable documents with your survey data (e.g., word file)

HOUSEHOLD MEMBER SURVEY FORM					
Household ID (assigned by Health Department)	Address	City	Zip Code	County	
\$ {hh_id}	\$ {address}	\$ {city}	\$ {zip}	\$ {county}	
Structure accessible for survey? \$ {structure_accessible}					
Primary Household Contact Name: \$ {hh_primary_name} Phone number(s): \$ {hh_primary_phone} # in Household: \$ {hh_total}					
Investigator Name(s): \$ {investigator}					
Survey Conducted in: \$ {survey_language} \$ {survey_language_other}					
No.	Household Member Demographics	Date/Time of Interview	Symptoms during Period of Interest (see above) and Onset Dates (if known)	Fevers (within 2 weeks before symptom onset, if household member symptomatic, or within past 2 months if asymptomatic)	Date/Time of Specimen Collection
\$ {n}, of \$ {hh_total} \$ {hh_age} \$ {hh_sex}	Last Name: \$ {hhn_lastname} First Name: \$ {hhn_firstname} Sex: \$ {hhn_sex} DOB: \$ {hhn_dob} \$ {hhn_dob_MM/YY} TIC: \$ {hhn_tic} \$ {hhn_age} = (2000 - \$ {hhn_dob_year}) \$ {hhn_gender} = (C=Male, F=Female) Onset date: \$ {hhn_onset_date} \$ {hhn_onset_date_MM/YY} (M/D/Y)	Date: \$ {hhn_interview_date} Time: \$ {hhn_interview_time} \$ {hhn_interview_date_MM/YY}	Fever: \$ {hhn_fever} (selected "yes") Yes: \$ {hhn_fever_selected_yes} No: \$ {hhn_fever_selected_no} Onset date: \$ {hhn_fever_onset_date} \$ {hhn_fever_onset_date_MM/YY} Rash: \$ {hhn_rash} (selected "yes") Yes: \$ {hhn_rash_selected_yes} No: \$ {hhn_rash_selected_no} Onset date: \$ {hhn_rash_onset_date} \$ {hhn_rash_onset_date_MM/YY} Red Eyes: \$ {hhn_red_eyes} (selected "yes") Yes: \$ {hhn_red_eyes_selected_yes} No: \$ {hhn_red_eyes_selected_no} Onset date: \$ {hhn_red_eyes_onset_date} \$ {hhn_red_eyes_onset_date_MM/YY} Joint Pain: \$ {hhn_joint_pain} (selected "yes") Yes: \$ {hhn_joint_pain_selected_yes} No: \$ {hhn_joint_pain_selected_no} Onset date: \$ {hhn_joint_pain_onset_date} \$ {hhn_joint_pain_onset_date_MM/YY} Seen at a hospital or clinic for symptoms?	\$ {hhn_fever} (selected "yes") Yes: \$ {hhn_fever_selected_yes} No: \$ {hhn_fever_selected_no} \$ {hhn_fever_selected_no} Location: \$ {hhn_location_selected} Dates of travel: From: \$ {hhn_travel_start_date} \$ {hhn_travel_start_date_MM/YY} To: \$ {hhn_travel_end_date} \$ {hhn_travel_end_date_MM/YY} \$ {hhn_travel_end_date}	Specimen: \$ {hhn_specimen} \$ {hhn_specimen_MM/YY} \$ {hhn_specimen} Specimen: \$ {hhn_specimen} \$ {hhn_specimen_MM/YY} \$ {hhn_specimen}

Make decision

○ Bring your data into ArcGIS

- ✓ Real-time data collection
 - Survey123 submitted data is immediately available for visualization and analysis
 - All data stored in ArcGIS feature services
- ✓ Use with other ArcGIS Apps
- ✓ Multiple Uses of data



○ Survy123 Examples

- ✓ Water quality study

<https://survey123.arcgis.com/share/55dc5e6e27294f3abb2fe220884476cf>



ArcGIS Collector

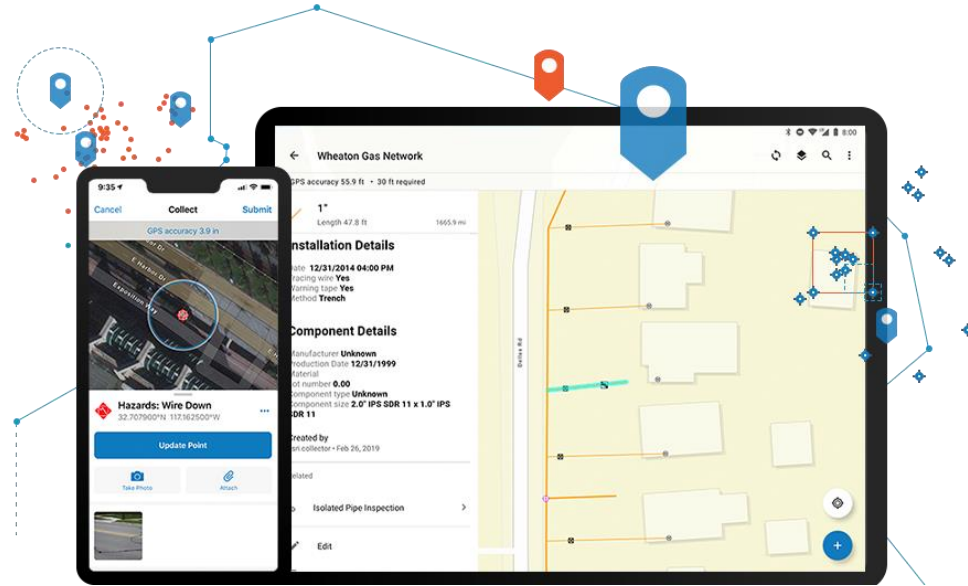
Overview

- ArcGIS Collector
 - ✓ Capture assets and observations
 - ✓ Perform inspections
- Workflow

Collector

- Map- centric data collection field solution

- ✓ Web maps
- ✓ Works offline
- ✓ High accuracy data



Collector

- Capture assets and observations
 - ✓ Improve accuracy and currency of data
 - ✓ Modernize field workflows



Hydrant
Collections



Sign
Inventory

Collector

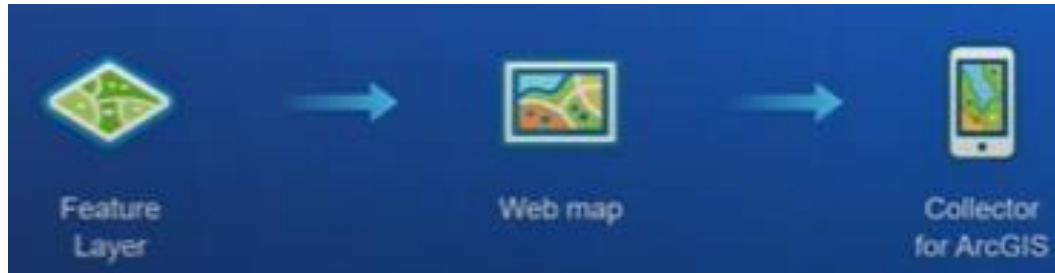
- Perform inspections
 - ✓ Replace redundant, inefficient field processes
 - ✓ Improve operational efficiency



Water valve inspections

Workflow

- Create feature layers to collect data
 - ✓ ArcGIS Online or ArcGIS Enterprise
- Add to a web map
 - ✓ Share the map and layers to a group
- Use the map in Collector



Workflow

- Create feature layers to collect data
 - ✓ Data sources
 - ArcGIS online hosted feature layer
 - ArcGIS Enterprise hosted feature layer
 - ArcGIS Server feature service (Enterprise geodatabase)
 - ✓ Create in ArcGIS Pro
 - Create new feature class and share as a feature layer
 - ✓ Create in ArcGIS Online or ArcGIS Enterprise with feature layer templates
 - Industry focused or Build your own layer
 - Completely customizable

○ Feature Layer Properties

✓ Configure editing capabilities

- Add, update, and delete (default)
- Update attributes and geometry (default)

✓ Operations

- Editor tracking
- Synchronization (for offline workflows) (default)

✓ Display

- Define style and rendering (Visualization tab)
- Define form (Data tab)

○ Feature Layer Properties

UCA Tree Mapping Collector_YH

OverviewDataVisualizationUsageSettings

Feature Layer (hosted)

Editing

☒ Enable editing.

This layer is shared with everyone. Enabling editing means everyone can edit it. If that isn't your intent, create a view of this layer that allows editing. [Learn more.](#)

☐ Keep track of created and updated features.

☒ Keep track of who created and last updated features.

☒ Enable Sync (required for offline use and collaboration).

• Who can edit features?

Share the layer to specific groups of people, the organization or publicly via the Share button on the Overview tab. This layer is currently shared with: Everyone (public)

• What kind of editing is allowed?

☒ Add

☒ Delete

☒ Update

☐ Attributes only

☒ Attributes and geometry

Workflow

○ Add to a Web Map

✓ Requirements

- At least one editable feature layer
- Basemap

✓ Map setting

- Search by layer
- Advanced offline settings and map areas

✓ Configure the form (pop-ups)

✓ Reference data

✓ Share with mobile workers via groups

○ Add to a Web Map

HomeGalleryMapSceneGroupsContentOrganization

UCA Tree Mapping Collector_YH

OverviewUsageSettings

GeneralOfflineWeb Map

Offline

Enabling offline mode allows this map to be downloaded and used in locations where there may be no internet connectivity. When offline mode is enabled, a user can download the map in apps that support offline workflows. [Learn how.](#)

Enable offline mode

Advanced Options

Map Areas

While not required, it is recommended that you create map areas ahead of time to package data for specific areas you expect to be used offline. Using map areas makes taking maps offline easier and faster.

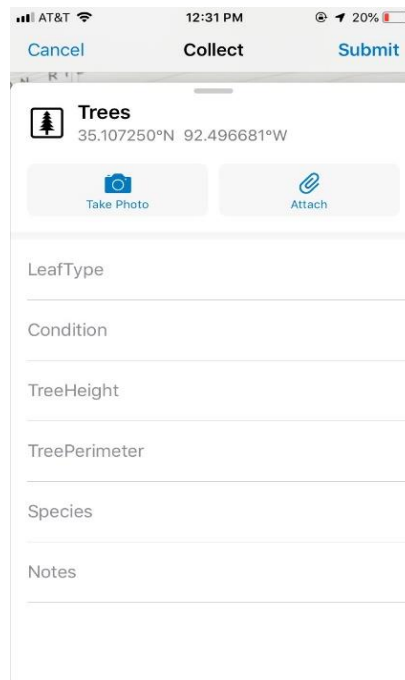
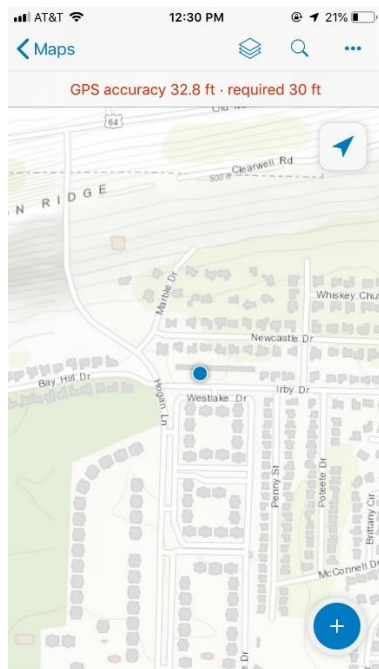
You have no map areas yet, click Manage Areas to get started.

Manage Areas

SaveCancel

Workflow

- Use the map in the Collector



-
- IOS: Collector (new generation) and Collector Classic
 - Android: Collector and Beta (new generation)

Thank you!